DEEP LEARNING MINI-PROJECT PROPOSAL

AI-Powered Real-Time Fruit Freshness & Shelf-Life Estimation

Computer Vision Based Automated Freshness Indexing & Quality Control

---

Domain: Deep Learning & Computer Vision | **Duration:** 2 Months (4 Phases) | **Target Accuracy:** 95%+

1. Executive Abstract

Project Summary for Academic Submission

Project Overview

Fruit quality assessment across agricultural supply chains and retail markets is heavily dependent on manual visual inspection, which is subjective, labor-intensive, and prone to human error. This project presents a high-precision, real-time Computer Vision system powered by Deep Learning to automate fruit classification and freshness evaluation.

Core Technology

Utilizing lightweight Convolutional Neural Networks (**MobileNetV2 / EfficientNet**) combined with **YOLOv8** object detection, the system identifies multiple fruit categories (Apple, Banana, Orange) and classifies them into *Fresh*, *Medium Fresh*, or *Stale/Rotten*.

Key Innovation

Beyond binary classification, the system calculates a **Freshness Quality Index (%)** and estimates the predicted **Remaining Shelf-Life (Days)**, deployed via an interactive Streamlit web interface with real-time camera support.

2. Problem Statement & Motivation

Why is this project relevant & high-impact?

Existing Challenges

- **High Post-Harvest Losses:** Over 30% of fruits spoil during storage & supply chain transit.
- **Subjective Quality Check:** Human inspection lacks consistency and is slow for mass processing.
- **Lack of Quantifiable Metrics:** Existing systems only mark "Good" or "Bad", without telling exact freshness percentage or remaining days.

30% - 40%

Global Fruit Wastage in Supply Chains

< 30 ms

Target Real-time Inference Latency

3. Deep Learning Architecture

Model Design & Computer Vision Pipeline

Component	Selected Framework / Model	Role & Functionality
Object Detection	YOLOv8 (Ultralytics)	Detects multiple fruits in live camera frame & draws bounding boxes.
Classification Model	MobileNetV2 / EfficientNetB0	Extracts deep features to classify fruit into Fresh vs Rotten.
Quality Indexing	Custom Softmax Regression	Outputs a normalized Freshness Quality Score (0% to 100%).
Shelf-Life Predictor	Regression Layer / Rule Engine	Estimates remaining edible days based on freshness grade & fruit type.
User Interface	Streamlit Web Dashboard	Interactive GUI with live webcam stream, gauge charts, and stats.

4. Project Roadmap: Month 1 (Phases 1 & 2)

15-Day Milestone Breakdown

Phase 1: Days 1 – 15 (Dataset & Preprocessing)

- Gather Kaggle Fresh/Rotten dataset (Apples, Bananas, Oranges).
- Clean dataset & remove corrupted/blurry images.
- Apply Data Augmentation (Rotation, Zoom, Brightness, Flipping).
- Split dataset: 80% Training, 10% Validation, 10% Testing.

Phase 2: Days 15 – 30 (Model Training & Fine-Tuning)

- Implement Transfer Learning with MobileNetV2 / ResNet50.
- Train model using Adam Optimizer & Categorical Cross-Entropy.
- Monitor validation loss and prevent overfitting using Early Stopping.
- Target Metrics: **92%+ Accuracy** & high F1-Score.

5. Project Roadmap: Month 2 (Phases 3 & 4)

15-Day Milestone Breakdown

Phase 3: Days 30 – 45 (UI & Feature Integration)

- Build interactive Streamlit Dashboard.
- Integrate Freshness Quality Index (%) gauge display.
- Develop Shelf-Life Prediction logic.
- Add single image upload & batch processing options.

Phase 4: Days 45 – 60 (Real-Time Camera & Deployment)

- Integrate OpenCV live webcam feed with YOLOv8.
- Test live detection under varying lighting conditions.
- Deploy web app on HuggingFace Spaces / Streamlit Cloud.
- Final documentation, project report, and demo preparation.

6. Requirements & Toolstack

Prerequisites for Execution

Category	Tool / Resource	Purpose / Usage
Compute Environment	Google Colab (T4 Free GPU)	Training heavy Deep Learning models without overloading laptop.
Development Software	Python 3.10+, VS Code / Jupyter	Primary programming language & IDE.
DL Frameworks	TensorFlow / PyTorch, Ultralytics YOLOv8	Model development, transfer learning & object detection.
Hardware Requirements	Laptop Webcam / Smartphone Cam	Capturing live video stream for real-time testing.
Physical Demonstration	Real Fruit Samples (Fresh & Aged)	For live presentation demo in front of evaluators.

7. Expected Deliverables & Future Scope

Project Impact & Applications

Final Deliverables

- Fully trained Deep Learning Model (`.h5` / `.onnx` weights).
- Interactive Web Dashboard with Live Webcam Feed.
- Complete Project Report, Phase Summaries & Presentation.
- Live Deployed Web Application Link.

Real-World Future Scope

- **Smart Refrigerators:** Automated food spoilage alerts.
- **Supermarket Automation:** Self-checkout quality inspection.
- **Agricultural Sorting:** Automated conveyor belt sorting in warehouses.